Primary Use Cases — Full Query-to-Response Traces

Each use case shows: where the query originates, what the trader types or clicks, the exact tool chain, where the LLM is involved, which Ghostfolio endpoints are called, and what comes back.

Use Case 1: "What game am I playing right now?"

Market Regime Check

Origin

The trader opens the agent dashboard Monday morning before market open. They want to know what kind of market they're walking into before deciding which strategies to run this week. They type into the chat interface or click a "Check Regime" button on the dashboard.

Example Query

"What's the market looking like this week?"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: regime_query
  → Context Check: no cached regime (start of day)

Tool Call 1: get_market_data(symbols=["SPY","^VIX","XLK","XLF","XLE","XLV","XLI","XLP","XLU","XLRE"],
period="60d")
  → yfinance returns OHLCV for all symbols
  → computes RSI, SMA(20,50,200), ATR, relative volume for each

Tool Call 2: detect_regime(index="SPY")
  → consumes get_market_data output (no separate fetch)
  → computes 5 dimensions:
    trend: SPY above rising 50 SMA, 20 SMA slope positive → trending_up
    volatility: VIX at 18 (40th percentile), ATR contracting → low_vol
    correlation: sector ETF avg pairwise correlation 0.62 → moderate
    breadth: 7/9 sector ETFs above 20 SMA → broad_participation
    rotation: XLK +3.2% 1wk vs XLU -0.8% → risk_on, tech leading
  → returns classification object with composite label + confidence

Synthesis (LLM #2)
  → "The market is in a trending, low-volatility environment with broad participation.
    This is historically the best regime for momentum and breakout strategies.
```

Tech is leading the rotation — your watchlist names in that sector have a tailwind. Mean reversion setups will fire less frequently in this regime but when they do, the snapback tends to be smaller. I'd prioritize breakout and momentum scans this week."

Verification (code)
→ checks: VIX=18 matches tool output ✓, 7/9 above 20 SMA matches ✓
→ confidence: 82% (4/5 dimensions clear, correlation is borderline)
→ no risk guardrails needed (informational query, no trade suggestion)

Ghostfolio Endpoints Called

None. This use case runs entirely on external market data (yfinance). Ghostfolio isn't needed for regime detection.

Response Shape

```json
{
  "summary": "Trending market, low volatility, broad participation...",
  "regime": {
    "trend": "trending_up",
    "volatility": "low_vol",
    "correlation": "moderate",
    "breadth": "broad_participation",
    "rotation": "risk_on_tech_leading",
    "composite": "trending_low_vol_broad",
    "confidence": 82
  },
  "strategy_implications": {
    "favored": ["momentum_continuation", "volatility_contraction_breakout"],
    "unfavored": ["mean_reversion"],
    "neutral": ["gap_fill", "post_earnings_drift"]
  },
  "disclaimer": "Market regime analysis, not financial advice."
}
```

LLM Calls: 2 (intent + synthesis)

Use Case 2: "Find me setups on my watchlist"

Opportunity Scanning

Origin

After checking the regime, the trader wants to see which stocks on their Ghostfolio watchlist match their active strategies. They type a query, or the agent proactively offers to scan after a regime check. Could also originate from a background cron scan that runs every 30-60 minutes during market hours and stores results as Signals.

Example Query

"Scan my watchlist for breakout and momentum setups"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: strategy_scan
  → Context Check: regime already cached from 5 min ago ✓ (skip re-fetch)

Tool Call 1: get_portfolio_snapshot()
  → Ghostfolio GET /portfolio/holdings → current positions
  → Ghostfolio GET /account → cash available
  → (needed so the agent knows what the trader already holds)

Tool Call 2: lookup via Ghostfolio GET /watchlist
  → returns: [NVDA, AMD, AAPL, MSFT, META, TSLA, AMZN, COIN, PLTR, GOOG, JPM, XOM]

Tool Call 3: get_market_data(symbols=watchlist, period="60d")
  → yfinance OHLCV for 12 symbols
  → computes all indicators

Tool Call 4: scan_strategies(strategy_id="volatility_contraction_breakout", universe=watchlist)
  → rules engine: ATR percentile < 25? price within 5% of 52wk high? volume declining?
  → matches: AAPL (score 84), MSFT (score 71)

Tool Call 5: scan_strategies(strategy_id="momentum_continuation", universe=watchlist)
  → rules engine: RSI 55-75? price > rising 20 EMA? volume above average?
  → matches: NVDA (score 88), META (score 76), PLTR (score 69)

Tool Call 6: check_risk(symbol="NVDA", direction="LONG", position_size_pct=5)
  → portfolio already holds 3% NVDA → would make 8% total → WARN: exceeds 5% limit
  → sector: adding more tech → tech already at 42% → WARN: sector concentration

Synthesis (LLM #2)
  → "I found 5 setups across your two strategies. NVDA is the strongest momentum
    signal (score 88) but you already hold 3% — adding more pushes you to 8% and
    your tech sector to 47%. If you want the cleanest risk-adjusted setup, AAPL's
```

volatility contraction breakout at 84 is strong and you don't have a current position. Entry near $237, stop at $229 (1.2 ATR below), target $254 (2.2:1 R:R). The regime supports both strategies this week."

Verification (code)
→ AAPL score 84 matches tool output ✓
→ NVDA 3% holding matches portfolio snapshot ✓
→ entry/stop/target math verified against ATR values ✓
→ risk warning for NVDA properly included ✓
→ confidence: AAPL 79% (strong signals + favorable regime), NVDA 88% (but risk issues)

Ghostfolio Endpoints Called

- `GET /portfolio/holdings` — current positions and allocation
- `GET /account` — cash balance and account info
- `GET /watchlist` — the user's saved watchlist symbols

Response Shape

```json
{
  "summary": "5 setups found across 2 strategies. AAPL is the cleanest...",
  "regime_context": "trending_low_vol_broad (favorable for both strategies)",
  "opportunities": [
    {
      "symbol": "NVDA", "strategy": "momentum_continuation", "score": 88,
      "direction": "LONG", "signals": ["RSI 62", "above rising 20 EMA", "volume 1.3x avg"],
      "entry": 142.50, "stop": 136.20, "target": 156.80, "risk_reward": "2.3:1",
      "warnings": ["Already hold 3%", "Tech sector at 42%"],
      "confidence": 88
    },
    {
      "symbol": "AAPL", "strategy": "volatility_contraction_breakout", "score": 84,
      "direction": "LONG", "signals": ["ATR 12th percentile", "within 2% of 52wk high", "volume declining 7 days"],
      "entry": 237.40, "stop": 229.80, "target": 254.20, "risk_reward": "2.2:1",
      "warnings": [],
      "confidence": 79
    }
  ],
  "disclaimer": "..."
}
```

LLM Calls: 2 (intent + synthesis)

Use Case 3: "Did I chart this right?"

Chart Validation

Origin

Sunday evening. The trader has spent an hour charting their watchlist for the week ahead. They've marked support, resistance, trendlines, and patterns on a stock. They open the agent and describe their analysis — either typing their levels or (post-MVP) uploading a screenshot. They want an honest second opinion before trading off these levels all week.

Example Query

"I'm looking at TSLA. I have support at $320, resistance at $365, and I see a bull flag forming. The uptrend line from the November low is still intact. Am I reading this right?"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: chart_validation
  → extracts: symbol=TSLA, user_support=320, user_resistance=365,
    user_pattern="bull flag", user_trendline="uptrend from November low"

Tool Call 1: lookup_symbol(query="TSLA")
  → Ghostfolio GET /symbol/lookup?query=TSLA
  → resolves to dataSource=YAHOO, symbol=TSLA

Tool Call 2: get_market_data(symbols=["TSLA"], period="6mo")
  → yfinance returns 6 months of daily OHLCV
  → computes SMA(20,50,200), RSI, ATR, Bollinger Bands, volume profile

Tool Call 3: detect_regime(index="SPY")
  → (if not cached) fetches and classifies current regime

Synthesis (LLM #2) — THIS IS THE KEY LLM REASONING TASK
  The LLM receives:
  - The trader's claimed levels and patterns
  - 6 months of OHLCV data with indicators
  - Current regime context

  The LLM reasons about:
```

- Support at $320: Was this tested? How many times? With what volume? What happened at each test? Is the level weakening or holding firm?
- Resistance at $365: Single touch or multiple rejections? Volume at the rejections? Is there a more significant resistance level nearby?
- Bull flag: Does the consolidation match flag criteria? How deep is the pullback relative to the prior move? Is volume declining in the flag?
- Uptrend line from November: How many touches? Any closes below it? Is the slope sustainable?
- What the trader MISSED: Any significant volume shelves? Gaps? Moving average confluence points? Weekly/monthly levels?
- Bias detection: Trader is clearly bullish (all levels support a long thesis). Are there bearish signals being ignored?

→ "Your $320 support is solid — tested 3 times since December with strong buying response each time. I'd narrow it to $318-322 as a zone rather than a single level. Your $365 resistance is weaker than you think — it's only been tested once in January, and that was a wick, not a close. The more significant resistance is $371-375 where there were two rejections with heavy volume in early January.

The bull flag is questionable. The pullback has retraced 58% of the prior move — classic flags retrace less than 50%. Volume has declined, which is good, but the depth makes me call this a consolidation rather than a flag. It could still break out but I'd want higher volume confirmation.

Your trendline from November is valid — 3 touches, no closes below. But it's running very close to the 50 SMA at $334, so if you're using the trendline as your stop reference, the 50 SMA break is probably the more meaningful level to watch.

One thing you didn't mark: there's a volume shelf at $340-345 from the December consolidation. About 30M shares traded in that range. It sits between your entry zone and your trendline — expect it to act as interim support or resistance."

Verification (code)
→ checks: $320 tested 3 times → verified against OHLCV lows near $318-322 ✓
→ 58% retracement → computed from flagpole high/low ✓
→ 50 SMA at $334 → matches indicator computation ✓
→ volume shelf at $340-345 → verified from volume-by-price analysis ✓

Ghostfolio Endpoints Called

- `GET /symbol/lookup?query=TSLA` — symbol resolution

Response Shape

```
json

{
  "summary": "Your $320 support is solid but resistance needs adjustment...",
  "level_validation": [
    { "type": "support", "user_level": 320, "agent_assessment": "valid",
      "detail": "3 tests since Dec, suggest $318-322 zone", "confidence": 85 },
    { "type": "resistance", "user_level": 365, "agent_assessment": "weak",
      "detail": "Single wick touch. Stronger resistance at $371-375", "confidence": 45 },
    { "type": "pattern", "user_pattern": "bull_flag", "agent_assessment": "questionable",
      "detail": "58% retracement too deep for classic flag", "confidence": 40 },
    { "type": "trendline", "user_trendline": "Nov uptrend", "agent_assessment": "valid",
      "detail": "3 touches, intact. Watch 50 SMA at $334 as more significant", "confidence": 78 }
  ],
  "missed_levels": [
    { "type": "volume_shelf", "level": "340-345", "detail": "~30M shares, Dec consolidation" }
  ],
  "bias_warning": "Analysis is entirely bullish. Consider: what if $320 fails?",
  "overall_chart_quality": 62,
  "disclaimer": "..."
}
```

LLM Calls: 2 (intent + synthesis) — but synthesis is heavier here (more reasoning tokens)

Use Case 4: "Why do I keep losing on breakouts?"

Trade Journal Intelligence

Origin

The trader has been using the agent for a few weeks. They've noticed their breakout trades keep failing and they want to understand why. This query comes from the chat interface, likely after a frustrating losing streak. It's the kind of question that Ghostfolio's existing portfolio tracking can't answer — Ghostfolio shows that you lost money, but not WHY.

Example Query

"My breakout trades have been terrible lately. What's going on?"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: journal_analysis
  → extracts: strategy_type="breakout", sentiment="frustrated", timeframe="recent"

Tool Call 1: get_trade_history(time_range="90d", strategy_tag="breakout")
  → Ghostfolio GET /order?range=90d&tags=breakout
  → returns 14 orders (7 BUY + 7 matching SELL entries)
  → enriches each with: P&L, hold duration, max adverse excursion, max favorable excursion

Tool Call 2: get_market_data(symbols=[symbols from trade history], period="90d")
  → fetches data for the 7 stocks traded
  → computes indicators at each entry date (what did the setup look like when they entered?)

Tool Call 3: detect_regime() — runs for each entry date in the history
  → classifies what regime was active when each trade was taken
  → result: 5 of 7 trades were entered during a ranging market

Tool Call 4: get_portfolio_snapshot()
  → current state for context

Synthesis (LLM #2) — BEHAVIORAL PATTERN RECOGNITION
  The LLM receives:
  - 7 breakout trades with full outcome data
  - Regime at time of each entry
  - Indicator data at each entry
  - Aggregate stats: 2 wins, 5 losses, win rate 29%, avg loss -3.2%, avg win +2.1%

  → "I see the problem. Five of your seven recent breakout trades were entered
     while the market was in a ranging regime. Your breakout strategy historically
     has a 65% win rate in trending markets but drops to 31% in ranging markets —
     and that's almost exactly what happened (29%).

     The market shifted from trending to ranging about 6 weeks ago, but you kept
     running the same playbook. This isn't a problem with your breakout criteria —
     when I look at the two wins, both were on days where the broader market showed
     temporary trending character within the range.

     There's a second issue: on 4 of the 5 losers, the stock hit your target zone
     within 3 days of your stop-out. Your stops are too tight for a ranging market
     where volatility expansion triggers false breakdowns. In a range, ATR is
     typically 30-40% higher than in a trend, so your standard 1.5 ATR stop is
```

getting clipped.

My recommendation: pause breakout scans until the regime detector shows trending again. Switch to mean reversion setups which are working well in this range. If you do take a breakout, widen stops to 2 ATR in ranging regimes."

Verification (code)
→ 5 of 7 in ranging regime → verified against regime classifications ✓
→ 29% win rate matches computed stats ✓
→ 4/5 losers hit target after stop → verified from max favorable excursion data ✓
→ ATR comparison trending vs ranging → verified from market data ✓

Ghostfolio Endpoints Called

- `GET /order` with filters — the trader's historical breakout-tagged orders
- `GET /portfolio/holdings` — current state

Response Shape

```json
{
  "summary": "5 of 7 breakout trades were in a ranging market where your win rate drops to 31%...",
  "analysis": {
    "total_trades": 7, "wins": 2, "losses": 5, "win_rate": "29%",
    "avg_win": "+2.1%", "avg_loss": "-3.2%", "profit_factor": 0.39,
    "root_cause": "regime_mismatch",
    "detail": "65% win rate trending, 31% ranging. Market shifted 6 weeks ago."
  },
  "secondary_finding": {
    "issue": "stops_too_tight",
    "detail": "4/5 losers hit target within 3 days of stop-out. ATR 30-40% higher in range."
  },
  "recommendations": [
    "Pause breakout scans until regime shifts to trending",
    "Switch to mean reversion (favorable in current range)",
    "If taking breakouts: widen stops to 2 ATR in ranging regimes"
  ],
  "disclaimer": "..."
}
```

LLM Calls: 2 (intent + synthesis) — synthesis is heavy (behavioral pattern reasoning)

Use Case 5: "Can I take this trade?"

Pre-Trade Risk Check

Origin

The trader has found a setup — either from the agent's scan, from their own analysis, or from a tip they saw elsewhere. Before entering, they want the agent to check it against their portfolio constraints and the current market context. This is the fastest, simplest use case — the trader wants a quick yes/no with specific reasons. Could also fire automatically when the trader clicks "Add to Ghostfolio" on an opportunity from Use Case 2.

Example Query

"I want to buy $8,000 of AMD. Does it fit?"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: risk_check
  → extracts: symbol=AMD, direction=LONG, amount=$8,000

Tool Call 1: lookup_symbol(query="AMD")
  → Ghostfolio GET /symbol/lookup?query=AMD
  → resolves dataSource + symbol

Tool Call 2: get_portfolio_snapshot()
  → Ghostfolio GET /portfolio/holdings → total value $156,000, holdings list
  → Ghostfolio GET /account → cash available $23,400
  → current AMD position: $4,200 (2.7%)
  → current tech sector: 38%

Tool Call 3: get_market_data(symbols=["AMD"], period="20d")
  → recent data for correlation check
  → also gets current price for position sizing math

Tool Call 4: check_risk(symbol="AMD", direction="LONG", position_size_pct=5.1)
  → $8,000 / $156,000 = 5.1% new position
  → existing AMD: 2.7% → total would be 7.8% → FAIL: exceeds 5% single position limit
  → tech sector: 38% + 5.1% = 43.1% → WARN: approaching 45% sector ceiling
  → correlation with existing NVDA position: 0.81 → WARN: high correlation
  → cash available: $23,400 → $8,000 is fine → PASS
  → overall: FAIL (position concentration)
```

```
Tool Call 5: detect_regime()
  → (cached from earlier) → ranging market
  → AMD is a momentum/breakout name → WARN: regime not ideal

Synthesis (LLM #2)
  → "This trade has issues. Adding $8,000 of AMD would put your AMD position at
    7.8% of your portfolio — above your 5% single-position limit. You'd also be at
    43% tech sector exposure, and AMD has a 0.81 correlation with your NVDA position,
    so you're effectively doubling down on the same bet.

    If you still like AMD here, I'd suggest sizing down to $3,600 (keeping total AMD
    at 5%) and acknowledging you're making a concentrated tech/semiconductor bet.

    One more thing: the regime is currently ranging, and AMD tends to be a momentum
    name. Your breakout trades have been struggling in this regime (29% win rate
    last 90 days). Make sure this entry has clear support levels and wider stops."

Verification (code)
  → 7.8% total AMD matches math ✓
  → 43% tech matches computation ✓
  → 0.81 correlation verified ✓
  → $3,600 adjusted size → 2.7% + 2.3% = 5.0% ✓
```

Ghostfolio Endpoints Called

- `GET /symbol/lookup?query=AMD` — symbol resolution
- `GET /portfolio/holdings` — current positions
- `GET /account` — cash balance

Response Shape

```
json
```

```
{
  "summary": "Trade has issues. AMD would be 7.8% of portfolio...",
  "risk_check": {
    "overall": "FAIL",
    "checks": [
      { "rule": "position_size", "status": "FAIL", "detail": "7.8% exceeds 5% limit" },
      { "rule": "sector_concentration", "status": "WARN", "detail": "Tech at 43.1%" },
      { "rule": "correlation", "status": "WARN", "detail": "0.81 with NVDA" },
      { "rule": "cash_available", "status": "PASS", "detail": "$23,400 available" }
    ]
  },
  "suggested_adjustment": {
    "max_position": "$3,600",
    "rationale": "Keeps total AMD at 5% limit"
  },
  "regime_warning": "Ranging market — AMD breakout trades at 29% win rate this regime",
  "disclaimer": "..."
}
```

LLM Calls: 2 (intent + synthesis)

Use Case 6: "What happened with SMCI?"

Signal Archaeology (Post-Move Analysis)

Origin

A stock just made a massive move — up 200% in a month, or crashed 50% in a week. The trader sees it and wonders "could I have caught that?" They ask the agent to look backward and identify what signals existed before the move. This is a learning/research query, not a trade execution query. Over time, the findings feed back into the strategy scanner to improve future detection.

Example Query

"SMCI ran 180% in the last 2 months. Was there anything that could have predicted it?"

Tool Chain

```
Intent Classification (LLM #1)
  → classifies as: signal_archaeology
  → extracts: symbol=SMCI, move="180% up", timeframe="2 months"


Tool Call 1: lookup_symbol(query="SMCI")
```

→ Ghostfolio GET /symbol/lookup?query=SMCI

Tool Call 2: get_market_data(symbols=["SMCI"], period="1y")
→ yfinance returns 1 year of OHLCV (need pre-move data for comparison)
→ computes all indicators across the full period

Tool Call 3: detect_regime() — for the period before the move started
→ what was the broad market doing when SMCI started running?

Tool Call 4: get_trade_history(symbol="SMCI")
→ Ghostfolio GET /order?symbol=SMCI
→ did the trader have any exposure? did they miss it entirely?

Synthesis (LLM #2) — FORENSIC ANALYSIS
The LLM receives full 1-year data and reasons backward:

→ "Looking at SMCI in the 4 weeks before the run started:

Volume: Trading volume increased 340% vs its 60-day average starting 3 weeks before the breakout. This was the earliest clear signal — unusual accumulation on no public news.

Technical: The stock had built a 6-week base between $38-42 with progressively tighter daily ranges (ATR dropped from $3.20 to $1.80). This is a textbook volatility contraction pattern — and it would have triggered your VCP strategy scanner at that point.

RSI: RSI held between 48-55 during the base — neutral, not overbought. This is the 'coiled spring' pattern where energy is building without extended momentum.

Relative strength: While the broader market was flat during this period, SMCI was quietly making higher lows. It was already outperforming before the move became obvious.

What your agent WOULD have caught with current tools: The VCP scanner would have flagged SMCI during the base formation. The volume anomaly tool (post-MVP) would have flagged the accumulation. The regime was trending at the time, which favors breakouts.

What would have required additional data sources (post-MVP): Social media mentions of SMCI increased 500% in the week before breakout. Unusual call option activity at the $45 strike appeared 2 weeks before the move.

Bottom line: with your current strategy library, you would have seen SMCI as a moderate-confidence VCP setup. The volume confirmation would have elevated it to high-confidence. The entry at the $42 breakout with a $38 stop would have captured the majority of the move."

Verification (code)
→ volume 340% increase verified against OHLCV data ✓
→ ATR decline from $3.20 to $1.80 verified ✓
→ RSI 48-55 range verified ✓
→ base range $38-42 verified ✓
→ regime at that date verified against stored classification ✓

Ghostfolio Endpoints Called

- `GET /symbol/lookup?query=SMCI` — symbol resolution
- `GET /order?symbol=SMCI` — any existing trades by the user

Response Shape

```
json
```

```
{
  "summary": "Multiple signals preceded the move, detectable with current tools...",
  "precursor_signals": [
    { "signal": "volume_accumulation", "timing": "3 weeks before",
      "strength": "strong", "detail": "340% above 60d avg", "detectable_now": true },
    { "signal": "volatility_contraction", "timing": "6 weeks before",
      "strength": "strong", "detail": "ATR $3.20→$1.80, textbook VCP", "detectable_now": true },
    { "signal": "relative_strength", "timing": "4 weeks before",
      "strength": "moderate", "detail": "Higher lows vs flat market", "detectable_now": true },
    { "signal": "social_mention_spike", "timing": "1 week before",
      "strength": "strong", "detail": "500% increase", "detectable_now": false,
      "requires": "social sentiment data provider (post-MVP)" },
    { "signal": "unusual_options_flow", "timing": "2 weeks before",
      "strength": "strong", "detail": "$45 strike call buying", "detectable_now": false,
      "requires": "options flow data provider (post-MVP)" }
  ],
  "hypothetical_trade": {
    "entry": 42.00, "stop": 38.00, "would_have_captured": "~160% of 180% move",
    "strategy_match": "volatility_contraction_breakout", "confidence_at_time": 74
  },
  "lesson": "VCP scanner + volume confirmation would have flagged this as high-confidence",
  "disclaimer": "Hindsight analysis. Past patterns don't guarantee future results."
}
```


LLM Calls: 2 (intent + synthesis) — synthesis is the heaviest of all use cases (forensic reasoning over a year of data)

Summary: Use Case → Tool Chain Map

Use Case	Origin	Tools Called	Ghostfolio Endpoints	LLM Calls
1. Regime Check	Dashboard, start of day	get_market_data → detect_regime	None	2
2. Opportunity Scan	After regime check, or cron	get_portfolio_snapshot → watchlist → get_market_data → scan_strategies (×N) → check_risk	/portfolio/holdings, /account, /watchlist	2
3. Chart Validation	Sunday prep, before trading week	lookup_symbol → get_market_data → detect_regime	/symbol/lookup	2 (heavy synthesis)

Use Case	Origin	Tools Called	Ghostfolio Endpoints	LLM Calls
4. Journal Analysis	After losing streak, performance review	get_trade_history → get_market_data → detect_regime (historical) → get_portfolio_snapshot	/order, /portfolio/holdings	2 (heavy synthesis)
5. Pre-Trade Risk Check	Before entering trade, or auto on "Add to Ghostfolio"	lookup_symbol → get_portfolio_snapshot → get_market_data → check_risk → detect_regime	/symbol/lookup, /portfolio/holdings, /account	2
6. Signal Archaeology	After big move, learning/research	lookup_symbol → get_market_data (1yr) → detect_regime (historical) → get_trade_history	/symbol/lookup, /order	2 (heaviest synthesis)